



KINROSS WOLAROI
SCHOOL

2022

Year 12 Mathematics Extension 2

Trial HSC Examination

Teacher Setting Paper: Mr Bowman

Head of Department: Mr Doyle

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculator may be used
- Write your answers for Section I on the multiple-choice answer sheet provided
- A reference sheet is provided at the back of this paper
- In Questions 11-16, show relevant mathematical reasoning and/or calculations.

Total marks –100

Section I – Multiple-Choice

10 marks

Attempt Questions 1-10

Allow 15 minutes for this section

Section II – Extended Response

90 marks

Attempt questions 11 – 16

Allow 2 hours and 45 minutes for this section

This examination paper does not necessarily reflect the content or format of the Higher School Certificate Examination in this subject.

Student Name/Number: _____

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Section I**10 marks****Attempt Questions 1 – 10****Allow about 15 minutes for this section**Use the multiple-choice answer sheet for Questions 1 - 10.**QUESTION 1**

Of the following, select the statement that is **TRUE** for all real numbers x and y :

- (A) If $x^3 > y^3$, then $x > y$
- (B) If $x^2 > y^2$, then $x > y$
- (C) If $x > y$, then $\frac{1}{x} < \frac{1}{y}$
- (D) If $|x| > |y|$, then $x > y$

QUESTION 2

Given that $|z - 2| = 2$ and $\arg(z - 2) = \frac{2\pi}{3}$, which of the following is an expression for z ?

- (A) $z = 1 + i\sqrt{3}$
- (B) $z = (2 + \sqrt{3}) + i$
- (C) $z = (2 - \sqrt{3}) + i$
- (D) $z = 3 + i\sqrt{3}$

QUESTION 3

$OABC$ is a rectangle with $\overrightarrow{OA} = 3\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$ and $\overrightarrow{OC} = 6\mathbf{i} + 4\mathbf{j} + a\mathbf{k}$ for some constant a . What is the value of a ?

- (A) -2
- (B) -3
- (C) -5
- (D) -6

QUESTION 4

Which of the following is an expression for $\int \frac{1}{\sqrt{3-2x-x^2}} dx$?

- (A) $\sin^{-1}\left(\frac{x-1}{2}\right) + C$
- (B) $\sin^{-1}\left(\frac{x+1}{2}\right) + C$
- (C) $\sin^{-1}\left(\frac{x-1}{\sqrt{2}}\right) + C$
- (D) $\sin^{-1}\left(\frac{x-1}{\sqrt{2}}\right) + C$

QUESTION 5

A particle is moving in a straight line with Simple Harmonic Motion. At any time t seconds it has displacement x metres from a fixed point on the line and velocity $v \text{ ms}^{-1}$ given by $v^2 = 9 - 4(x-1)^2$. What is the amplitude of the motion?

- (A) 1.5 metres
- (B) 2 metres
- (C) 2.5 metres
- (D) 3 metres

QUESTION 6

Consider the statement

For any function $f(x)$, $f(x)$ is not continuous at $x = c \Rightarrow f(x)$ is not differentiable at $x = c$.

Which of the following is correct?

- (A) The contrapositive statement is false and the converse statement is false.
- (B) The contrapositive statement is false and the converse statement is true.
- (C) The contrapositive statement is true and the converse statement is false.
- (D) The contrapositive statement is true and the converse statement is true.

QUESTION 7

Which of the following is an expression for $e^{i3\theta} + e^{i\theta}$?

- (A) $2\sin\theta e^{i2\theta}$
- (B) $2\cos\theta e^{i2\theta}$
- (C) $2\sin 2\theta e^{i\theta}$
- (D) $2\cos 2\theta e^{i\theta}$

QUESTION 8

The points A, B, C are collinear where $\overrightarrow{OA} = \mathbf{i} - \mathbf{j}$, $\overrightarrow{OB} = -3\mathbf{j} - \mathbf{k}$ and $\overrightarrow{OC} = 2\mathbf{i} + a\mathbf{j} + b\mathbf{k}$ for some constants a and b . What are the values of a and b ?

- (A) $a = -1$ and $b = -1$
- (B) $a = -1$ and $b = 1$
- (C) $a = 1$ and $b = -1$
- (D) $a = 1$ and $b = 1$

QUESTION 9

If $\int_1^4 f(x)dx = k$ what is the value of $\int_1^4 f(5-x)dx$?

- (A) $-k$
- (B) $5 - k$
- (C) $k + 5$
- (D) k

QUESTION 10

A stone is projected from a point O on flat ground with speed $V\sqrt{2} \text{ ms}^{-1}$ at an angle 45° above the horizontal. The stone moves in a vertical plane under gravity where the acceleration due to gravity is $g \text{ ms}^{-2}$. Air resistance may be neglected.

At time t seconds the position vector of the stone relative to O is $\mathbf{r}(t) = Vt\mathbf{i} + \left(Vt - \frac{1}{2}gt^2\right)\mathbf{j}$.

Which one of the following about the trajectory of the stone is correct?

- (A) Its horizontal range is 2 times its maximum height.
- (B) Its horizontal range is 3 times its maximum height.
- (C) Its horizontal range is 4 times its maximum height.
- (D) Its horizontal range is 5 times its maximum height.

END OF SECTION I

Section II**90 marks****Attempt Questions 11 – 16****Allow about 2 hours 45 minutes for this section**

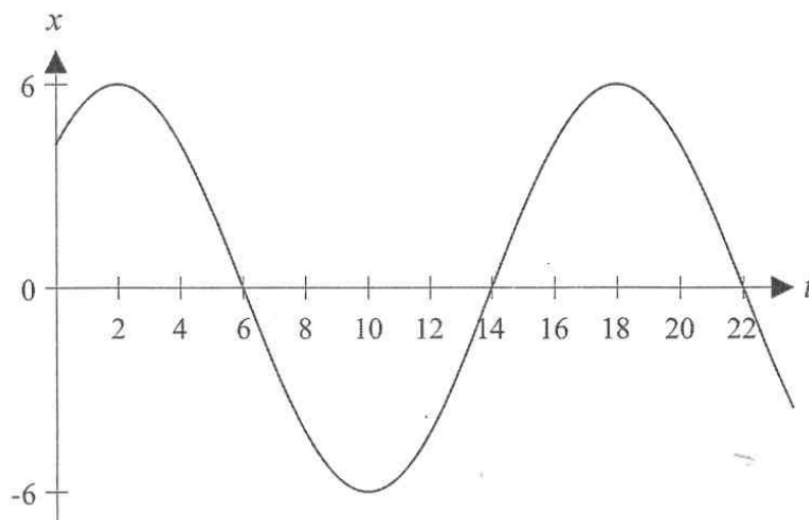
Answer each question in the booklets provided. Extra writing booklets are available.

Marks**QUESTION 11 (15 marks) Start a new writing booklet**

- (a) The complex numbers z and w are represented on the Argand diagram by the points Z and W respectively. $z = 2 + 2i\sqrt{3}$ and the point W is obtained by rotating the point Z in a clockwise direction about the origin through an angle of 90° .
- (i) Find z and w in modulus-argument form. **2**
- (ii) Find zw and $\frac{z}{w}$ in modulus-argument form. **2**
- (b) (i) Find $\int \frac{x}{\sqrt{1+3x^2}} dx$. **2**
- (ii) **3**
Evaluate $\int_0^{\frac{\pi}{4}} \tan x \, dx$.
- (c) Consider the equation $z^3 - 11z^2 + 55z - 125 = 0$.
- (i) Find the three roots of the equation in the form $a + ib$, where a, b are real. **3**
- (ii) Show that the points A, B and C in the Argand diagram representing these roots **3**
lie on a circle of the form $|z| = k$ for some constant k , and find the area of $\triangle ABC$.

QUESTION 12 (15 marks) Start a new writing booklet

(a)



The graph shows the displacement x cm from the centre of motion at time t seconds for a particle performing Simple Harmonic Motion in a straight line.

- (i) Find an expression for x as a function of t . 2
- (ii) Find the distance travelled by the particle in the first **minute** of its motion after observation began at time $t = 0$. 2

(b)

- (i) Find real numbers a, b and c such that 3

$$\frac{1}{x^2(x-1)} = \frac{a}{x} + \frac{b}{x^2} + \frac{c}{x-1}.$$

- (ii) Hence find 2

$$\int \frac{1}{x^2(x-1)} dx$$

(c)

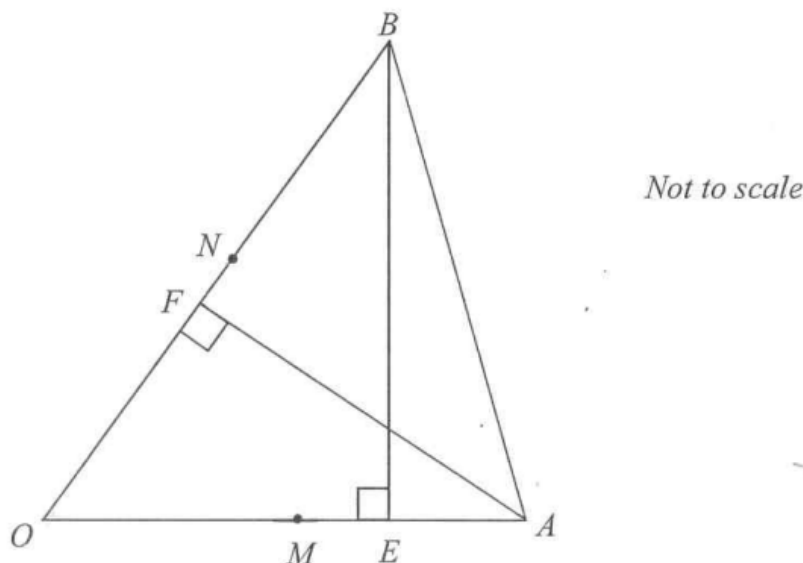
- If $I_n = \int_1^e (1 - \ln x)^n dx$ for $n = 0, 1, 2, \dots$ 3
- show that $I_n = -1 + nI_{n-1}$

- (d) Use Mathematical Induction to prove that 3

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n} \text{ for } n \geq 2$$

QUESTION 13 (15 marks) Start a new writing booklet

(a)



In $\triangle OAB$, BE is the altitude from B to OA and AF is the altitude from A to OB .

3

M, N are the midpoints of OA, OB respectively. $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$.

Use vector methods to show that $|OM| \times |OE| = |ON| \times |OF|$.

- (b) A body of mass m kg is travelling in a horizontal straight line so that the resultant force on the body is a resistance force of magnitude $\frac{1}{10}m\sqrt{1+v}$ when its speed is $v \text{ ms}^{-1}$. Initially the speed of the body is 15 ms^{-1} .

(i) Find the time taken for the body to come to rest.

3

(ii) Find the distance travelled by the body in coming to rest.

3

- (c) Consider the lines L_1, L_2 determined by the vector equations

$$L_1 : \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \text{ and } L_2 : \mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

Show that L_1 and L_2 intersect and are perpendicular, stating the coordinates of the point of intersection.

3

- (d) (i) Find the parametric equations of the line through $(-2, 1, 2)$ that is parallel to $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$.

2

(i) Determine whether either of the points $(1, 2, 3)$ or $(-6, 3, -4)$ lie on this line.

1

QUESTION 14 (15 marks) Start a new writing booklet

- (a) (i) Use de Moivre's Theorem to show that 3

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$$

- (ii) Solve the equation $\cos 5\theta = -1$ for $0 \leq \theta \leq 2\pi$. 4

Hence show that

$$\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$$

and that

$$\cos \frac{\pi}{5} \cos \frac{3\pi}{5} = -\frac{1}{4}.$$

- (b) $P(6, 3, -4)$, $Q(3, 1, 1)$ and $R(2, -1, 3)$ are the vertices of a triangle. 4

Find the size of the largest angle (to the nearest minute).

- (c) (i) Show that 2

$$\frac{n}{n+1} {}^{2n}C_n = {}^{2n}C_{n-1} \text{ for } n \geq 1$$

- (ii) Hence show that 2

$$\frac{1}{n+1} {}^{2n}C_n \text{ is an integer for } n \geq 1$$

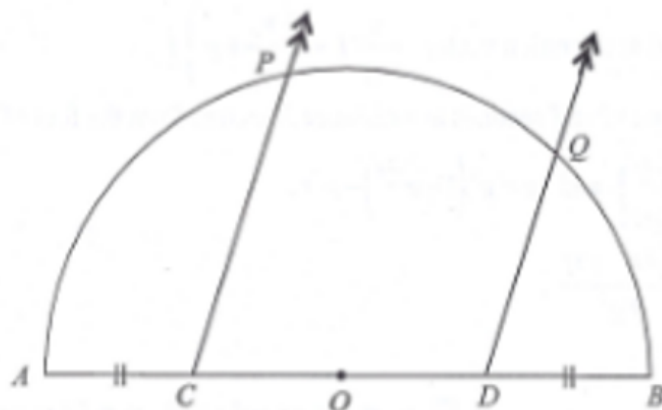
QUESTION 15 (15 marks) Start a new writing booklet

- (a) Use the substitution $t = \tan \frac{x}{2}$ to find in simplest exact form the value of

4

$$\int_0^{\frac{\pi}{2}} \frac{1}{3 + 5\sin x} dx$$

(b)



A semicircle is drawn on diameter AB . O is the midpoint of AB and points C and D lie on AB such that $AC = BD$. Parallel lines are drawn through C and D intersecting the semicircle at P and Q respectively. $\vec{OC} = \tilde{c}$ and $\vec{CP} = \tilde{p}$.

- (i) Explain why $\vec{DQ} = \lambda \tilde{p}$ for some scalar $\lambda > 0$

1

- (ii) Hence show that $(1 - \lambda)\tilde{p} + 2\tilde{c} = 0$.

3

- (c) Prove by contradiction that $\log_5 7$ is irrational

3

- (d) A particle moves in a straight line. At time t seconds the particle has a displacement of x m, a velocity of v ms^{-1} and acceleration a ms^{-2} . Initially the particle has displacement 0 m and velocity 2 ms^{-1} . The acceleration is given by $a = -2e^{-x}$. The velocity of the particle is always positive.

- (i) Show that $v = 2e^{-\frac{x}{2}}$.

2

- (ii) Find an expression for x as a function of t .

2

QUESTION 16 (15 marks) Start a new writing booklet

(a)

- (i) Show that, for all positive values of x and y , 1

$$\frac{x+y}{2} \geq \sqrt{xy}$$

- (ii) Hence prove that 2

$$(b+c)(c+a)(a+b) \geq 8abc$$

- (b) The displacement of a particle at time t is x , measured from a fixed point O where a, c are positive constants. If 5

$$\frac{dx}{dt} = a(c^2 - x^2)$$

and $x = 0$ when $t = 0$, prove that

$$x = \frac{c(e^{2act} - 1)}{e^{2act} + 1}.$$

If $x = 3$ when $t = 1$, and $x = \frac{75}{17}$ when $t = 2$, show that $c = 5$ and evaluate a .

- (c) Find an expression for 4

$$\int \sec^3 \theta \, d\theta$$

- (d) If p, q and r are positive real numbers and $p + q \geq r$, prove that 3

$$\frac{p}{1+p} + \frac{q}{1+q} - \frac{r}{1+r} \geq 0$$

END OF PAPER.



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MULTIPLE-CHOICE ANSWER SHEET

For multiple choice questions, choose the best answer A, B, C or D and fill in the correct circle.

1. ☐ A ☐ B ☐ C ☐ D
2. ☐ A ☐ B ☐ C ☐ D
3. ☐ A ☐ B ☐ C ☐ D
4. ☐ A ☐ B ☐ C ☐ D
5. ☐ A ☐ B ☐ C ☐ D
6. ☐ A ☐ B ☐ C ☐ D
7. ☐ A ☐ B ☐ C ☐ D
8. ☐ A ☐ B ☐ C ☐ D
9. ☐ A ☐ B ☐ C ☐ D
10. ☐ A ☐ B ☐ C ☐ D



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AACBA CBDDC

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- (A) -2
- (B) -3
- (C) -5
- (D) -6

QUESTION 4

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Consider the statement

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The points A, B, C are collinear where $\overrightarrow{OA} = \mathbf{i} - \mathbf{j}$, $\overrightarrow{OB} = -3\mathbf{j} - \mathbf{k}$ and $\overrightarrow{OC} = 2\mathbf{i} + a\mathbf{j} + b\mathbf{k}$ for some constants a and b . What are the values of a and b ?

- (A) $a = -1$ and $b = -1$
- (B) $a = -1$ and $b = 1$
- (C) $a = 1$ and $b = -1$
- (D) $a = 1$ and $b = 1$

QUESTION 9

If $\int_1^4 f(x)dx = k$ what is the value of $\int_1^4 f(5-x)dx$?

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Which one of the following about the trajectory of the stone is correct?

- (A) Its horizontal range is 2 times its maximum height.
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- (C) Its horizontal range is 4 times its maximum height.
- (D) Its horizontal range is 5 times its maximum height.

END OF SECTION I

MULTIPLE CHOICE DETAILED SOLUTIONS

Question	Answer	Solution	Outcomes
1	A	Function needs to be monotonic increasing or decreasing so that if $a \geq b$ then $f(a) \geq f(b)$. Only option A satisfies this requirement.	MEX12-2
2	A	$z - 2 = 2\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right) = -1 + i\sqrt{3} \quad \therefore z = 1 + i\sqrt{3}$	MEX12-4
3	C	$\angle AOC = 90^\circ \quad \therefore (3\hat{i} - 2\hat{j} + 2\hat{k}) \cdot (6\hat{i} + 4\hat{j} + a\hat{k}) = 0 \quad \therefore a = -5$	MEX12-3
4	B	$\int \frac{1}{\sqrt{3-2x-x^2}} dx = \int \frac{1}{\sqrt{4-(x+1)^2}} dx = \sin^{-1}\left(\frac{x+1}{2}\right) + c$	MEX12-5
5	A	$v^2 = 9 - 4(x-1)^2 \quad \therefore v = 0$ for $x-1 = \pm \frac{3}{2}$, $x = -\frac{1}{2}$ or $x = \frac{5}{2}$. Particle moves 3 m between its extreme positions. Amplitude is 1.5 m.	MEX12-6
6	C	Contrapositive: $f(x)$ is differentiable at $x = c \Rightarrow f(x)$ is continuous at $x = c$ True Converse: $f(x)$ is not differentiable at $x = c \Rightarrow f(x)$ is not continuous at $x = c$ False	MEX12-2
7	B	$e^{i3\theta} + e^{i\theta} = (1 + e^{i2\theta})e^{i\theta}$ $(1 + e^{i2\theta}) = 1 + \cos 2\theta + i \sin 2\theta = 2 \cos \theta (\cos \theta + i \sin \theta) = 2 \cos \theta e^{i\theta}$ $\therefore e^{i3\theta} + e^{i\theta} = 2 \cos \theta e^{i2\theta}$	MEX12-4
8	D	A, B, C collinear $\Leftrightarrow \overrightarrow{BA} = \lambda \overrightarrow{AC}$ for some scalar λ . $\overrightarrow{BA} = \hat{i} + 2\hat{j} + \hat{k} \quad \overrightarrow{AC} = \hat{i} + (a+1)\hat{j} + b\hat{k} \quad \therefore a = 1, b = 1$	MEX12-3
9	D	Substituting $u = 5 - x$, $\int_1^4 f(5-x) dx = \int_4^1 f(u)(-1) du = \int_1^4 f(u) du = k$	MEX12-5
10	C	$\dot{y} = 0 \Rightarrow V - gt = 0 \quad \therefore t = \frac{V}{g}$ and $y = \frac{V^2}{g}\left(1 - \frac{1}{2}\right) = \frac{V^2}{2g}$ at maximum height. $y = 0 \Rightarrow Vt - \frac{1}{2}gt^2 = 0 \quad \therefore t = \frac{2V}{g}$ and $x = \frac{2V^2}{g}$ at horizontal range. $\frac{2V^2}{g} = 4 \times \frac{V^2}{2g}$. Hence horizontal range is 4 times maximum height.	MEX12-6

QUESTION 11

(a) (i)

$$z = 4\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = 4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) \quad w = 4\left(\cos\left(\frac{\pi}{3} - \frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{3} - \frac{\pi}{2}\right)\right) = 4\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$$

(a) (ii)

$$zw = 4 \times 4 \left\{ \cos\left(\frac{\pi}{3} + \left(-\frac{\pi}{6}\right)\right) + i\sin\left(\frac{\pi}{3} + \left(-\frac{\pi}{6}\right)\right) \right\} = 16\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

$$\frac{z}{w} = \frac{4}{4} \left\{ \cos\left(\frac{\pi}{3} - \left(-\frac{\pi}{6}\right)\right) + i\sin\left(\frac{\pi}{3} - \left(-\frac{\pi}{6}\right)\right) \right\} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$$

(b) (i)

$$\begin{aligned} & \int \frac{x}{\sqrt{1+3x^2}} dx \\ &= \frac{1}{6} \int 6x(1+3x^2)^{-\frac{1}{2}} dx \\ &= \frac{1}{6} (1+3x^2)^{\frac{1}{2}} \times 2 + C \\ &= \frac{1}{3} \sqrt{1+3x^2} + C \end{aligned}$$

(b)(ii)

$$\begin{aligned} & \int_0^{\frac{\pi}{4}} \tan x dx \\ &= [\ln \cos x]_0^{\frac{\pi}{4}} \\ &= \ln 1 - \ln\left(\frac{1}{\sqrt{2}}\right) \\ &= \frac{1}{2} \ln 2 \end{aligned}$$

(c) (i)

$$z^3 - 11z^2 + 55z - 125 = 0$$

$P(5) = 0$ hence $(z - 5)$ is a factor

$$z^3 - 11z^2 + 55z - 125 = (z - 5)(z^2 + az + 25)$$

Equating terms in z^2 :

$$-11 = a - 5$$

Hence

$$a = -6$$

$$(z - 5)(z^2 - 6z + 25) = 0$$

$$z = 5, z = 3 \pm 4i$$

QUESTION 12

(a)(i)

Amplitude is 6 cm. Period is 16 s. $\therefore \frac{2\pi}{n} = 16 \quad \therefore n = \frac{\pi}{8}$.

Graph is a translation 2 units to the right of $x = 6\cos\frac{\pi}{8}t$. $\therefore x = 6\cos\frac{\pi}{8}(t-2)$

(a)(ii)

$60 = 16 \times 3 + 8 + 4$ Hence 60 seconds is the time for 3 oscillations, plus one half an oscillation, plus twice the time taken to travel from its initial position to its position at time $t = 2$.

$t = 0 \Rightarrow x = 6\cos\left(-\frac{\pi}{4}\right) = 3\sqrt{2}$. Hence particle travels $(6 - 3\sqrt{2})$ cm in first 2 seconds.

The distance travelled in $3\frac{1}{2}$ oscillations is $3\frac{1}{2} \times 24$ cm = 84 cm.

Hence distance travelled in the first minute is $84 + 2(6 - 3\sqrt{2})$ cm = $(96 - 6\sqrt{2})$ cm

(b)(i)

$$\frac{1}{x^2(x-1)} = \frac{a}{x} + \frac{b}{x^2} + \frac{c}{x-1}.$$
$$1 = ax(x-1) + b(x-1) + cx^2$$

Subst $x = 1$

$$1 = c$$

Equate coeffs of x^2

$$a + c = 0$$

$$a = -1$$

Equate constants:

$$1 = -b$$

$$b = -1$$

$$\frac{1}{x^2(x-1)} = \frac{-1}{x} - \frac{1}{x^2} + \frac{1}{x-1}$$

(b)(ii)

$$\int \frac{1}{x^2(x-1)} dx$$
$$= \int -\frac{1}{x} - \frac{1}{x^2} + \frac{1}{x-1} dx$$
$$= -\ln x + \frac{1}{x} + \ln(x-1) + C$$
$$= \frac{1}{x} + \ln\left(\frac{x-1}{x}\right) + C$$

(c)

$$\begin{aligned} I_n &= \int_1^e (1 - \ln x)^n dx \\ &= \left[x(1 - \ln x)^n \right]_1^e - \int_1^e nx(1 - \ln x)^{n-1} \left(-\frac{1}{x} \right) dx, \quad n=1, 2, 3, \dots \\ &= -1 + n \int_1^e (1 - \ln x)^{n-1} dx \\ &= -1 + nI_{n-1} \end{aligned}$$

(d) To prove that

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n} \text{ for } n \geq 2$$

$$\begin{aligned} n &= 2 \\ LHS &= 1 - \frac{1}{2^2} = \frac{3}{4} \\ RHS &= \frac{3}{4} \end{aligned}$$

Hence true for $n = 2$

Assume true for $n = k$, i.e. assume that

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{k^2}\right) = \frac{k+1}{2k} \dots \dots \dots (A)$$

To prove true for $n = k + 1$ i.e. that

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{k^2}\right) \left(1 - \frac{1}{(k+1)^2}\right) = \frac{k+2}{2(k+1)} \dots \dots \dots (B)$$

LHS of equation (B)

$$\begin{aligned} &= \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{k^2}\right) \left(1 - \frac{1}{(k+1)^2}\right) \\ &= \left(\frac{k+1}{2k}\right) \left(1 - \frac{1}{(k+1)^2}\right) \text{ using assumption (A)} \\ &= \left(\frac{k+1}{2k}\right) \left(\frac{(k+1)^2 - 1}{(k+1)^2}\right) \\ &= \left(\frac{k+1}{2k}\right) \left(\frac{k^2 + 2k}{(k+1)^2}\right) \\ &= \left(\frac{k+1}{2k}\right) \left(\frac{k(k+2)}{(k+1)^2}\right) \\ &= \frac{k+2}{2(k+1)} \text{ as required} \end{aligned}$$

Hence if true for $n = k$, then true for $n = k + 1$.

True for $n = 2$, therefore true for all integral $n \geq 2$.

QUESTION 13

(a)

$\vec{OM} = \frac{1}{2}\vec{a}$ and $\vec{OE} = \lambda\vec{a}$ for some scalar λ . Then $|\vec{OM}| |\vec{OE}| = \frac{1}{2}\lambda \vec{a} \cdot \vec{a}$

$\vec{ON} = \frac{1}{2}\vec{b}$ and $\vec{OF} = \mu\vec{b}$ for some scalar μ . Then $|\vec{ON}| |\vec{OF}| = \frac{1}{2}\mu \vec{b} \cdot \vec{b}$

$\vec{AF} \perp \vec{OB} \quad \therefore (\mu\vec{b} - \vec{a}) \cdot \vec{b} = 0 \quad \therefore \mu \vec{b} \cdot \vec{b} = \vec{a} \cdot \vec{b}$

Similarly $\vec{BE} \perp \vec{OA}$ gives $\lambda \vec{a} \cdot \vec{a} = \vec{b} \cdot \vec{a}$. But $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$. Hence $|\vec{OM}| |\vec{OE}| = |\vec{ON}| |\vec{OF}| = \frac{1}{2} \vec{a} \cdot \vec{b}$

(b)(i)

Let the body come to rest in T seconds

$$\begin{aligned} \ddot{x} &= -\frac{1}{10}\sqrt{1+v} \\ \frac{dv}{dt} &= -\frac{1}{10}\sqrt{1+v} \\ \int_{15}^0 \frac{1}{\sqrt{1+v}} dv &= -\frac{1}{10} \int_0^T dt \\ 2\left[\sqrt{1+v}\right]_{15}^0 &= -\frac{1}{10}T \\ 2(1-4) &= -\frac{1}{10}T \\ T &= 60 \end{aligned}$$

Hence body comes to rest in 1 minute.

(b)(ii)

Let the body come to rest after travelling X metres.

$$\begin{aligned} v \frac{dv}{dx} &= -\frac{1}{10}\sqrt{1+v} \\ \int_{15}^0 \frac{v}{\sqrt{1+v}} dv &= -\frac{1}{10} \int_0^X dx \\ \int_{15}^0 \left(\sqrt{1+v} - \frac{1}{\sqrt{1+v}} \right) dv &= -\frac{1}{10}X \\ \left[\frac{2}{3}(1+v)^{\frac{3}{2}} - 2\sqrt{1+v} \right]_{15}^0 &= -\frac{1}{10}X \\ \frac{2}{3}(1-64) - 2(1-4) &= -\frac{1}{10}X \\ X &= 360 \end{aligned}$$

Hence body travels 360 m in coming to rest.

(c)

$$3 + 2\lambda = -1 + \mu \quad (1)$$

At any intersection point $2 - \lambda = 1 + \mu \quad (2)$ is a set of 3 consistent simultaneous equations.

$$-1 + \lambda = -\mu \quad (3)$$

Considering (1) and (2) : $(1) - (2) \Rightarrow 1 + 3\lambda = -2$ and hence $\lambda = -1, \mu = 2$.

Substituting in (3) : LHS = -2 = RHS

Hence the set of 3 equations is consistent with solution $\lambda = -1, \mu = 2$.

$$\text{At intersection } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}. \quad \text{Also } \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = 2 - 1 - 1 = 0.$$

Hence the lines intersect at right angles at $(1, 3, -2)$

(d)(i)

$$\frac{x+2}{2} = \frac{y-1}{-1} = \frac{z-2}{3} = \lambda$$

Hence

$$x = 2\lambda - 2, y = 1 - \lambda, z = 3\lambda + 2$$

(d)(ii)

Substituting $(1, 2, 3)$ gives

$$1 = 2\lambda - 2, 2 = 1 - \lambda, 3 = 3\lambda + 2$$
$$\lambda = \frac{3}{2}, -1, \frac{1}{3}$$

λ is not the same hence point $(1, 2, 3)$ does **NOT** lie on line

Substituting $(-6, 3, -4)$ gives

$$-6 = 2\lambda - 2, 3 = 1 - \lambda, -4 = 3\lambda + 2$$
$$\lambda = -2, -2, -2$$

λ is the same hence point $(-6, 3, -4)$ **DOES** lie on line.

QUESTION 14

(a)(i)

$$\begin{aligned}
 \cos 5\theta + i \sin 5\theta &= (\cos \theta + i \sin \theta)^5 \\
 \therefore \cos 5\theta &= \operatorname{Re}(\cos \theta + i \sin \theta)^5 \\
 &= {}^5C_0 \cos^5 \theta + {}^5C_2 \cos^3 \theta (i \sin \theta)^2 + {}^5C_4 \cos \theta (i \sin \theta)^4 \\
 &= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2 \\
 &= \cos^5 \theta (1 + 10 + 5) + \cos^3 \theta (-10 - 10) + 5 \cos \theta \\
 &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta
 \end{aligned}$$

(a)(ii)

$$\cos 5\theta = -1 \text{ for } 5\theta = n\pi, n = 1, 3, 5, \dots \therefore \theta = \frac{\pi}{5}, \frac{3\pi}{5}, \pi, \frac{7\pi}{5}, \frac{9\pi}{5} \text{ for } 0 \leq \theta \leq 2\pi.$$

$$\text{Hence } 16x^5 - 20x^3 + 5x + 1 = 0 \text{ has roots } \cos \frac{\pi}{5}, \cos \frac{3\pi}{5}, -1, \cos \frac{7\pi}{5}, \cos \frac{9\pi}{5}.$$

$$\text{But } \cos \frac{7\pi}{5} = \cos \left(2\pi - \frac{3\pi}{5}\right) = \cos \frac{3\pi}{5} \text{ and similarly } \cos \frac{9\pi}{5} = \cos \frac{\pi}{5}$$

Hence using the relationships between coefficients and roots :

$$\begin{aligned}
 2 \cos \frac{\pi}{5} + 2 \cos \frac{3\pi}{5} - 1 &= 0 & -\left(\cos \frac{\pi}{5} \cos \frac{3\pi}{5}\right)^2 &= -\frac{1}{16} \\
 \therefore \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} &= \frac{1}{2} & \therefore \cos \frac{\pi}{5} \cos \frac{3\pi}{5} &= \pm \frac{1}{4} \\
 & & \text{But } \cos \frac{\pi}{5} > 0 \text{ and } \cos \frac{3\pi}{5} < 0 & \\
 & & \therefore \cos \frac{\pi}{5} \cos \frac{3\pi}{5} &= -\frac{1}{4}
 \end{aligned}$$

(b)

$$P(6, 3, -4), Q(3, 1, 1) \text{ and } R(2, -1, 3)$$

Largest angle is opposite largest side.

Side lengths:

$$PQ^2 = 38$$

$$QR^2 = 9$$

$$PR^2 = 81$$

PR longest side hence angle PQR is required angle.

$$\begin{aligned}
 \overrightarrow{PQ} &= -3\mathbf{i} - 2\mathbf{j} + 5\mathbf{k} \text{ so } PQ = \sqrt{38} \\
 \overrightarrow{QR} &= -\mathbf{i} - 2\mathbf{j} + 2\mathbf{k} \text{ so } QR = 3
 \end{aligned}$$

$$\begin{aligned}
 \cos \angle PQR &= \frac{3^2 + 38 - 81}{2 \times 3 \times \sqrt{38}} \\
 &= -\frac{17}{3\sqrt{38}} \\
 \theta &= 156^\circ 49'
 \end{aligned}$$

(c)(i)

$$\frac{n}{n+1} {}^{2n}C_n = \frac{n(2n)!}{(n+1)n!n!} = \frac{(2n)!}{(n+1)!(n-1)!} = {}^{2n}C_{n-1}$$

(c)(ii)

$$\frac{1}{n+1} {}^{2n}C_n = \left(1 - \frac{n}{n+1}\right) {}^{2n}C_n = {}^{2n}C_n - {}^{2n}C_{n-1}$$

Since these binomial coefficients are respectively the number of ways of choosing n or $(n-1)$ items from $2n$ items, both are integers hence their difference is also an integer.

QUESTION 15

(a)

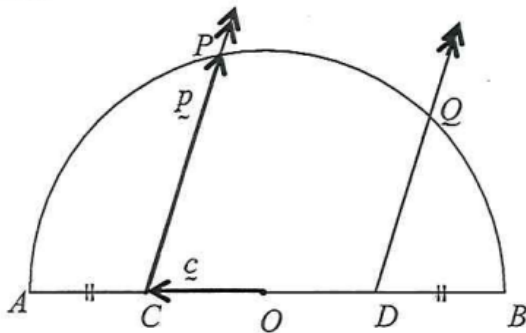
$$\begin{aligned}
 t &= \tan \frac{x}{2} \\
 dt &= \frac{1}{2} \sec^2 \frac{x}{2} dx \\
 \frac{2}{1+t^2} dt &= dx \\
 x=0 &\Rightarrow t=0 \\
 x=\frac{\pi}{2} &\Rightarrow t=1
 \end{aligned}$$

$$\begin{aligned}
 3+5\sin x &= \frac{3(1+t^2)+10t}{1+t^2} \\
 &= \frac{(3t+1)(t+3)}{1+t^2} \\
 \frac{1}{3+5\sin x} &= \left(\frac{1+t^2}{2} \right) \left\{ \frac{3}{3t+1} - \frac{1}{t+1} \right\}
 \end{aligned}$$

$$\begin{aligned}
 \int_0^{\frac{\pi}{2}} \frac{1}{3+5\sin x} dx &= \int_0^1 \left\{ \frac{3}{3t+1} - \frac{1}{t+1} \right\} dt \\
 &= \left[\ln \frac{3t+1}{t+1} \right]_0^1 \\
 &= \ln 2 - \ln 1 \\
 &= \ln 2
 \end{aligned}$$

(b)(i) & (ii)

Answer



$$\begin{aligned}
 \vec{DQ}, \vec{CP} &\text{ are parallel in the same direction.} \\
 \text{Hence } \vec{DQ} &= \lambda \vec{p} \text{ for some } \lambda > 0. \\
 \text{Then } \vec{OP} &= \vec{c} + \vec{p} \text{ and } \vec{OQ} = -\vec{c} + \lambda \vec{p} \\
 \text{But } |\vec{OP}| &= |\vec{OQ}| \text{ (radii of the circle)} \\
 \therefore (\vec{c} + \vec{p}) \cdot (\vec{c} + \vec{p}) &= (-\vec{c} + \lambda \vec{p}) \cdot (-\vec{c} + \lambda \vec{p}) \\
 \vec{c} \cdot \vec{c} + \vec{p} \cdot \vec{p} + 2\vec{c} \cdot \vec{p} &= \vec{c} \cdot \vec{c} + \lambda^2 \vec{p} \cdot \vec{p} - 2\lambda \vec{c} \cdot \vec{p} \\
 (1 - \lambda^2) \vec{p} \cdot \vec{p} + 2(1 + \lambda) \vec{c} \cdot \vec{p} &= 0 \\
 (1 - \lambda) \vec{p} \cdot \vec{p} + 2\vec{c} \cdot \vec{p} &= 0
 \end{aligned}$$

(c.) Let $\log_5 7 = p$ and assume this is rational, i.e. of form $\frac{x}{y}$.

$$\begin{aligned}
 7 &= 5^{\frac{x}{y}} \\
 7^y &= 5^x
 \end{aligned}$$

Which is impossible as all powers of 7 end in a 1, 3, 7 or 9 whereas all powers of 5 end in a 5.
Hence $\log_5 7$ is irrational

(d)(i)

$$\begin{aligned}
 a &= -2e^{-x} \\
 v \cdot \frac{dv}{dx} &= -2e^{-x} \\
 v dv &= -2e^{-x} dx \\
 \int_2^v v dv &= - \int_0^x 2e^{-x} dx \\
 \frac{v^2}{2} - 2 &= 2e^{-x} - 2 \\
 v^2 &= 4e^{-x} \\
 v &= 2e^{-\frac{x}{2}}
 \end{aligned}$$

(d)(ii)

$$v = 2e^{-\frac{x}{2}}$$

$$\frac{dx}{dt} = 2e^{-\frac{x}{2}}$$

$$\frac{1}{2}e^{\frac{x}{2}}dx = dt$$

$$\int_0^x \frac{1}{2}e^{\frac{x}{2}}dx = \int_0^t dt$$

$$e^{\frac{x}{2}} - 1 = t$$

$$e^{\frac{x}{2}} = t + 1$$

$$\frac{x}{2} = \ln(t + 1)$$

$$x = 2\ln(t + 1)$$

QUESTION 16

(a)(i)

$$\begin{aligned} \text{since } (x - y)^2 &\geq 0 \\ \therefore x^2 + y^2 &\geq 2xy \\ x^2 + 2xy + y^2 &\geq 4xy \\ (x + y)^2 &\geq 4xy \\ (x + y) &\geq 2\sqrt{xy} \\ \frac{x + y}{2} &\geq \sqrt{xy} \text{ as required.} \end{aligned}$$

(a) (ii)

To prove $(b + c)(c + a)(a + b) \geq 8abc$

From (a)(i),

$$\begin{aligned} \frac{b + c}{2} &\geq \sqrt{bc} \\ \text{i.e. } b + c &\geq 2\sqrt{bc} \end{aligned}$$

Similarly,

$$\begin{aligned} c + a &\geq 2\sqrt{ca} \\ a + b &\geq 2\sqrt{ab} \end{aligned}$$

Hence

$$\begin{aligned} (b + c)(c + a)(a + b) &\geq 8\sqrt{a^2b^2c^2} \\ \text{i.e. } (b + c)(c + a)(a + b) &\geq 8abc \text{ as required.} \end{aligned}$$

(b)

$$\frac{dx}{dt} = a(c^2 - x^2)$$

$$\frac{dt}{dx} = \frac{1}{a} \cdot \frac{1}{c^2 - x^2}$$

Integrating,

$$\begin{aligned} t &= \frac{1}{2ac} \ln \left(\frac{c + x}{c - x} \right) + C \\ t &= 0 \text{ when } x = 0 \\ C &= 0 \\ t &= \frac{1}{2ac} \ln \left(\frac{c + x}{c - x} \right) \\ \frac{c + x}{c - x} &= e^{2act} \\ c + x &= (c - x)e^{2act} \\ c(2e^{act} - 1) &= x(e^{2act} + 1) \\ x &= \frac{c(e^{2act} - 1)}{e^{2act} + 1} \text{ as required.} \end{aligned}$$

$$x = 3 \text{ when } t = 1$$

$$3 = \frac{c(e^{2ac} - 1)}{e^{2ac} + 1} \dots (i)$$

$$x = \frac{75}{17} \text{ when } t = 2$$

$$\frac{75}{17} = \frac{c(e^{4ac} - 1)}{e^{4ac} + 1} \dots (ii)$$

From (i),

$$ce^{2ac} - c = 3e^{2ac} + 3$$

$$e^{2ac} = \frac{c+3}{c-3} \dots (iii)$$

From (ii)

$$75e^{4ac} + 75 = 17ce^{4ac} - 17c$$

$$e^{4ac} = \frac{17c+75}{17c-75} \dots (iv)$$

Since

$$e^{4ac} = (e^{2ac})^2 \text{ from (iii) and (iv) we have}$$

$$\frac{17c+75}{17c-75} = \left(\frac{c+3}{c-3}\right)^2$$

$$(c^2 - 6c + 9)(17c + 75) = (c^2 + 6c + 9)(17c - 75)$$

$$17c^3 + 75c^2 - 102c^2 - 450c + 153c + 675 = 17c^3 - 75c^2 + 102c^2 - 450c + 153c - 675$$

$$54c^2 - 1350 = 0$$

$$c = 5$$

$$e^{10a} = 4$$

$$10a = \ln 4$$

$$a = \frac{1}{5} \ln 2$$

(c)

$$\text{Let } I = \int \sec^3 x \, dx$$

$$= \int \sec x \cdot \sec^2 x \, dx$$

$$= \sec x \tan x - \int \sec x \tan x \cdot \tan x \, dx$$

$$= \sec x \tan x - \int \sec x (\sec^2 x - 1) \, dx$$

$$= \sec x \tan x - \int \sec^3 x \, dx + \int \sec x \, dx$$

$$I = \sec x \tan x - I + \ln(\sec x + \tan x) + C$$

$$2I = \sec x \tan x + \ln(\sec x + \tan x) + C$$

$$\int \sec^3 x \, dx = \frac{1}{2} (\sec x \tan x + \ln[\sec x + \tan x]) + C$$

(d)

To prove $\frac{p}{1+p} + \frac{q}{1+q} - \frac{r}{1+r} \geq 0$ given $p + q \geq r$

(This is one of many ways)

$$\begin{aligned}\frac{p}{1+p} + \frac{q}{1+q} - \frac{r}{1+r} &= \frac{p(1+q)(1+r) + q(1+p)(1+r) - r(1+p)(1+q)}{(1+p)(1+q)(1+r)} \\&= \frac{p + pr + pq + pqr + q + qr + pq + pqr - r - rq - rp - pqr}{(1+p)(1+q)(1+r)} \\&= \frac{p + 2pq + q - r}{(1+p)(1+q)(1+r)} \\&\geq \frac{r + 2pq - r}{(1+p)(1+q)(1+r)} \quad \text{since } p + q \geq r \\&= \frac{2pq}{(1+p)(1+q)(1+r)} \\&\geq 0 \text{ since } p, q, r > 0 \text{ (given)}\end{aligned}$$

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours 45 minutes for this section

Answer each question in the booklets provided. Extra writing booklets are available.

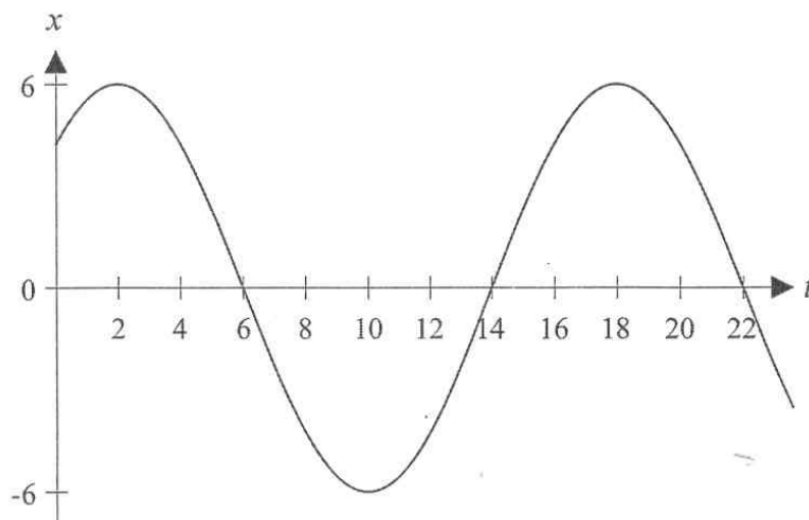
Marks

QUESTION 11 (15 marks) Start a new writing booklet

- (a) The complex numbers z and w are represented on the Argand diagram by the points Z and W respectively. $z = 2 + 2i\sqrt{3}$ and the point W is obtained by rotating the point Z in a clockwise direction about the origin through an angle of 90° .
- (i) Find z and w in modulus-argument form. 2
- (ii) Find zw and $\frac{z}{w}$ in modulus-argument form. 2
- (b) (i) Find $\int \frac{x}{\sqrt{1+3x^2}} dx$. 2
- (ii) Evaluate $\int_0^{\frac{\pi}{4}} \tan x \, dx$. 3
- (c) Consider the equation $z^3 - 11z^2 + 55z - 125 = 0$.
- (i) Find the three roots of the equation in the form $a + ib$, where a, b are real. 3
- (ii) Show that the points A, B and C in the Argand diagram representing these roots lie on a circle of the form $|z| = k$ for some constant k , and find the area of $\triangle ABC$. 3

QUESTION 12 (15 marks) Start a new writing booklet

(a)



The graph shows the displacement x cm from the centre of motion at time t seconds for a particle performing Simple Harmonic Motion in a straight line.

- (i) Find an expression for x as a function of t . 2
- (ii) Find the distance travelled by the particle in the first **minute** of its motion after observation began at time $t = 0$. 2

(b)

- (i) Find real numbers a, b and c such that 3

$$\frac{1}{x^2(x-1)} = \frac{a}{x} + \frac{b}{x^2} + \frac{c}{x-1}.$$

- (ii) Hence find 2

$$\int \frac{1}{x^2(x-1)} dx$$

(c)

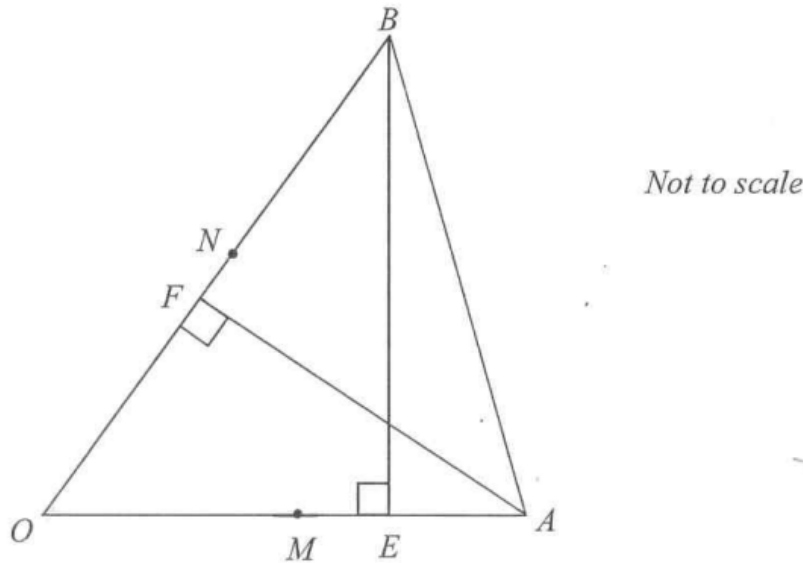
- If $I_n = \int_1^e (1 - \ln x)^n dx$ for $n = 0, 1, 2, \dots$ 3
- show that $I_n = -1 + nI_{n-1}$

- (d) Use Mathematical Induction to prove that 3

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n} \text{ for } n \geq 2$$

QUESTION 13 (15 marks) Start a new writing booklet

(a)



In $\triangle OAB$, BE is the altitude from B to OA and AF is the altitude from A to OB .

3

M, N are the midpoints of OA, OB respectively. $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$.

Use vector methods to show that $|OM| \times |OE| = |ON| \times |OF|$.

- (b) A body of mass m kg is travelling in a horizontal straight line so that the resultant force on the body is a resistance force of magnitude $\frac{1}{10}m\sqrt{1+v}$ when its speed is $v \text{ ms}^{-1}$. Initially the speed of the body is 15 ms^{-1} .

(i) Find the time taken for the body to come to rest.

3

(ii) Find the distance travelled by the body in coming to rest.

3

- (c) Consider the lines L_1, L_2 determined by the vector equations

$$L_1 : \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \text{ and } L_2 : \mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

Show that L_1 and L_2 intersect and are perpendicular, stating the coordinates of the point of intersection.

3

- (d) (i) Find the parametric equations of the line through $(-2, 1, 2)$ that is parallel to $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$.

2

(i) Determine whether either of the points $(1, 2, 3)$ or $(-6, 3, -4)$ lie on this line.

1

QUESTION 14 (15 marks) Start a new writing booklet

- (a) (i) Use de Moivre's Theorem to show that 3

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$$

- (ii) Solve the equation $\cos 5\theta = -1$ for $0 \leq \theta \leq 2\pi$. 4

Hence show that

$$\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$$

and that

$$\cos \frac{\pi}{5} \cos \frac{3\pi}{5} = -\frac{1}{4}.$$

- (b) $P(6, 3, -4)$, $Q(3, 1, 1)$ and $R(2, -1, 3)$ are the vertices of a triangle. 4

Find the size of the largest angle (to the nearest minute).

- (c) (i) Show that 2

$$\frac{n}{n+1} {}^{2n}C_n = {}^{2n}C_{n-1} \text{ for } n \geq 1$$

- (ii) Hence show that 2

$$\frac{1}{n+1} {}^{2n}C_n \text{ is an integer for } n \geq 1$$

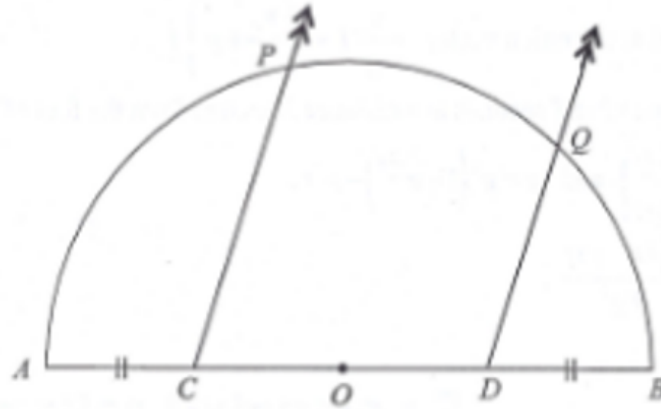
QUESTION 15 (15 marks) Start a new writing booklet

- (a) Use the substitution $t = \tan \frac{x}{2}$ to find in simplest exact form the value of

4

$$\int_0^{\frac{\pi}{2}} \frac{1}{3 + 5\sin x} dx$$

(b)



A semicircle is drawn on diameter AB . O is the midpoint of AB and points C and D lie on AB such that $AC = BD$. Parallel lines are drawn through C and D intersecting the semicircle at P and Q respectively. $\overrightarrow{OC} = \tilde{c}$ and $\overrightarrow{CP} = \tilde{p}$.

- (i) Explain why $\overrightarrow{DQ} = \lambda \tilde{p}$ for some scalar $\lambda > 0$

1

- (ii) Hence show that $(1 - \lambda)\tilde{p} + 2\tilde{c} = 0$.

3

- (c) Prove by contradiction that $\log_5 7$ is irrational

3

- (d) A particle moves in a straight line. At time t seconds the particle has a displacement of x m, a velocity of v ms^{-1} and acceleration a ms^{-2} . Initially the particle has displacement 0 m and velocity 2 ms^{-1} . The acceleration is given by $a = -2e^{-x}$. The velocity of the particle is always positive.

- (i) Show that $v = 2e^{-\frac{x}{2}}$.

2

- (ii) Find an expression for x as a function of t .

2

QUESTION 16 (15 marks) Start a new writing booklet

(a)

- (i) Show that, for all positive values of x and y , 1

$$\frac{x+y}{2} \geq \sqrt{xy}$$

- (ii) Hence prove that 2

$$(b+c)(c+a)(a+b) \geq 8abc$$

- (b) The displacement of a particle at time t is x , measured from a fixed point O where a, c are positive constants. If 5

$$\frac{dx}{dt} = a(c^2 - x^2)$$

and $x = 0$ when $t = 0$, prove that

$$x = \frac{c(e^{2act} - 1)}{e^{2act} + 1}.$$

If $x = 3$ when $t = 1$, and $x = \frac{75}{17}$ when $t = 2$, show that $c = 5$ and evaluate a .

- (c) Find an expression for 4

$$\int \sec^3 \theta \, d\theta$$

- (d) If p, q and r are positive real numbers and $p + q \geq r$, prove that 3

$$\frac{p}{1+p} + \frac{q}{1+q} - \frac{r}{1+r} \geq 0$$

END OF PAPER.



2022

Year 12 Mathematics Extension 2

Trial HSC Examination

MULTIPLE-CHOICE ANSWER SHEET

For multiple choice questions, choose the best answer A, B, C or D and fill in the correct circle.

1. ☐ A ☐ B ☐ C ☐ D
2. ☐ A ☐ B ☐ C ☐ D
3. ☐ A ☐ B ☐ C ☐ D
4. ☐ A ☐ B ☐ C ☐ D
5. ☐ A ☐ B ☐ C ☐ D
6. ☐ A ☐ B ☐ C ☐ D
7. ☐ A ☐ B ☐ C ☐ D
8. ☐ A ☐ B ☐ C ☐ D
9. ☐ A ☐ B ☐ C ☐ D
10. ☐ A ☐ B ☐ C ☐ D